AI × Web3 Developers Challenge

Official Problem Statement & Technical Briefing

Organised by **French Tech Lagos**
As part of the **French Week** program led by **FNCCI**
Date: Tuesday, 3 November 2026
Venue: Mike Adenuga Centre, Ikoyi, Lagos
Challenge theme: Resilient Households, Smarter Savings

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1. Welcome and Introduction

Welcome to the AI × Web3 Developers Challenge, organised by French Tech Lagos as part of the French Week program.

Today, developers will work collaboratively to explore how emerging technologies can address a practical challenge affecting households: maintaining consistent savings habits.

The objective is not simply to build another financial application. It is to demonstrate how AI and Web3 can empower households to navigate financial uncertainty, build consistent savings habits and achieve their financial goals.

You will work with an existing technical foundation and are expected to build a functional solution within the allocated time.

2. The Challenge Scenario

David and Amara are a young Nigerian couple.

They have a combined monthly income of ₦X and want to save ₦2 million over 12 months towards an important life goal.

They have regular expenses such as NEPA/electricity bills, housekeeping, transportation, food and other household expenses. They also face occasional unexpected expenses and have different spending and saving patterns.

Both want to contribute to their shared goal, but maintaining consistency is difficult. Some months they save less than planned; in others, unexpected expenses force them to pause or reduce their contributions.

They don't need another place simply to keep their money.

They need a smarter way to plan, stay motivated and remain consistent while building their savings over time.

3. The Problem Statement

How might we help David and Amara build and maintain consistent savings habits using Artificial Intelligence and Web3?

Your task is to design and develop a solution that helps the couple:

- Understand their household financial position and savings behaviour.
- Set and manage a shared savings goal.
- Track contributions and monitor progress over time.
- Identify spending patterns and potential barriers to consistent saving.
- Receive useful insights and recommendations that encourage better savings decisions.
- Maintain transparent, verifiable records of their savings commitments and recorded contributions using blockchain technology.

Think beyond a basic savings tracker. Focus on the experience, intelligence and mechanisms that can help users turn savings intentions into consistent habits.

4. What You Are Expected to Build

Develop a functional prototype of a smart household savings solution.

Your solution should demonstrate the following capabilities:

A. Household financial dashboard

Present a clear overview of the household's financial position, including relevant income, expenses, current savings balance and savings goal.

B. Savings goal management

Allow users to view and manage their shared savings goal, including:

- Target amount: ₦2 million.
- Target period: 12 months.
- Current savings balance.
- Amount remaining.
- Progress towards the goal.

C. Savings activity and history

Display savings contributions over time. Help users understand their savings patterns and how consistently they are progressing towards their target.

D. AI-powered financial insights

Use the supplied AI analysis endpoint to present insights based on the household's financial information.

Your interface should make the insights understandable and actionable. Examples include:

- Savings consistency observations.
- Spending patterns.
- Progress towards the savings target.
- Recommendations for improving savings behaviour.
- Suggested next actions.

E. Web3 and blockchain integration

Connect to the supplied SavingsGoal smart contract deployed on Celo Sepolia.

Demonstrate how users can:

- Connect a compatible wallet.
- Create or register a savings goal on the smart contract.
- Retrieve their goal information.
- Record a contribution value.
- Display the contribution total recorded on-chain.

Important: The smart contract records numerical contribution values. It does not transfer, hold or verify real money. This is a testnet demonstration, not a real financial transaction.

F. User experience

Build an intuitive and accessible experience. The solution should make it easy for a household to understand its progress, interpret insights and take meaningful action.

5. Technical Resources and Existing Integrations

A functional technical boilerplate will be provided to all participating teams. It already includes the foundational integrations required for the challenge.

Participants are expected to explore, understand, extend and build upon the existing implementation, rather than recreate the integrations.

A. Frontend

- React
- Vite

The frontend boilerplate is already connected to the supplied Apidog mock APIs.

The following endpoints are available:

Method	Endpoint	Purpose
GET	/customer	Household/customer information
GET	/transactions	Household transactions
GET	/savings	Savings balance and contribution history
GET	/goal	Savings goal information
POST	/ai/analyse-savings	Savings analysis and recommendations

B. AI Integration

The AI analysis workflow is already integrated into the frontend boilerplate and connected to the supplied analysis endpoint.

For the challenge, AI API keys will be provided to enable live AI-powered savings analysis, with the existing mock analysis response serving as a fallback.

Participants should focus on:

- Exploring the existing AI integration and understanding its architecture.
- Using the provided AI credentials and available integration.
- Generating insights based on household income, expenses and savings behaviour.
- Presenting personalised observations and actionable recommendations.
- Ensuring the application remains functional when live AI analysis is unavailable by falling back to the predefined mock response.

Important:

- No personal AI API keys or subscriptions are required.
- Use only the AI credentials and configuration provided by the coaches.
- Do not expose API keys in frontend source code or public GitHub repositories.
- The mock response must remain available as a fallback.

C. Web3 and Wallet Integration

The wallet integration is already included in the boilerplate, together with the connection to the deployed SavingsGoal smart contract.

Network configuration:

- Network: Celo Sepolia Testnet
- Chain ID: 11142220
- RPC: https://forno.celo-sepolia.celo-testnet.org
- Contract address: 0xf3bF2289BB778bE70829DC4E7C630db4B348f5DE
- Explorer: https://celo-sepolia.blockscout.com

The existing smart contract supports:

Function	Purpose
createGoal()	Create or replace a user's savings goal

getGoal()	Retrieve the connected user's goal
recordContribution()	Record a numerical contribution value

Participants should focus on extending the existing wallet and smart contract functionality and incorporating blockchain interactions into their product experience.

D. Development Environment

Participants should have access to:

- Provided GitHub boilerplate repository.
- VS Code or another preferred code editor.
- Node.js and npm.
- A compatible browser and internet connection.
- MetaMask or another compatible wallet.

6. Important Technical Clarifications

The frontend is already connected to the Apidog mock APIs.

The AI integration is already included. Coaches will provide AI API credentials for live analysis, with a predefined mock response as fallback.

Wallet connectivity and smart contract integration are already included.

The smart contract operates on Celo Sepolia testnet. No real money is transferred or held.

Blockchain contribution records represent numerical values and do not verify actual bank deposits or payments.

Use only synthetic household and financial data.

Keep API credentials and private keys out of source code and public repositories.

7. Suggested Development Approach

- Explore the existing boilerplate and understand its architecture.
- Review the existing API data and household financial information.

- Build the household financial dashboard and savings management experience.
- Extend the existing AI integration to generate useful savings insights using the provided credentials.
- Ensure the mock analysis response works as a fallback.
- Improve the wallet and smart contract interaction experience.
- Test the complete user journey and prepare the final demonstration.

You are free to make product and design decisions as long as they address the problem statement and demonstrate the required capabilities.

8. Submission Requirements

Before the development sprint ends, each team must submit:

- A GitHub repository containing the solution.
- A working application or demonstrable prototype.
- A brief explanation of the solution and the problem it addresses.
- A demonstration of the AI analysis integration.
- A demonstration of the Web3/smart contract integration.
- A short explanation of important technical and product decisions.

Developers should ensure their repositories are accessible to the technical jury.

9. Evaluation Process

The challenge will have two evaluation stages.

Stage 1: Technical Evaluation and Pre-selection

The technical jury will assess submissions with support from the coaches.

The evaluation will consider:

Evaluation criterion	Weight
Technical functionality	20
Architecture and code quality	15
AI integration	15
Web3 and blockchain implementation	20
System integration and data handling	15
Engineering execution	15

Total	**100**

This stage will identify the teams that demonstrate the strongest technical implementation and are ready for the final presentation.

Stage 2: Final Jury Evaluation

Shortlisted teams will present their solutions to the general jury.

The final evaluation will consider:

Evaluation criterion	Weight
Problem understanding and relevance	20
Innovation and originality	20
Practical value and impact	25
Product experience and usability	15
Presentation and communication	20
Total	**100**

The general jury will assess the overall value, relevance and quality of the proposed solutions.

10. Roles and Expectations

Developers

- Work and manage time effectively.
- Build on the provided technical resources.
- Ensure the solution addresses the stated problem.
- Submit a working prototype and prepare a clear demonstration.

Coaches

- Support teams with technical guidance and problem-solving.
- Help participants understand the provided resources and integrations.
- Encourage good engineering practices.
- Guide rather than build the solution on behalf of participants.

Technical Jury

- Assess technical implementation and integration.
- Review the submitted repositories and demonstrations.

- Identify and recommend teams for the final stage.

General Jury

- Evaluate shortlisted solutions based on relevance, innovation, usability, impact and presentation.
- Participate in the final deliberation and selection of winners.

11. Challenge Guidelines

- Respect the allocated development time.
- Use the provided boilerplate, APIs and smart contract.
- Do not use real financial data or attempt real-money transactions.
- Do not expose private keys, wallet recovery phrases or other credentials.
- Keep your implementation focused on the problem statement.
- Be prepared to explain and demonstrate your work.

12. Closing

This challenge is an opportunity to demonstrate how technology can move beyond features and solve meaningful everyday problems.

We encourage you to think critically, collaborate, experiment and build with purpose.

The goal is not simply to demonstrate AI or blockchain technology. It is to show how these technologies can work together to create a practical, accessible and meaningful solution.

From Intention to Habit. Let's build!

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